SECTION – I: GENERAL

1. SCOPE:

1.1 This specification covers design, engineering, manufacture, assembly, stage testing, inspection and testing before supply and delivery at site F.O.R. destination of outdoor equipment/materials in EHT transmission line / substations in the State of Andhra Pradesh.

1.2 It is not the intent to specify completely herein all details of the design and construction of equipment. However, the equipment shall conform in all respects to high standards of engineering, design and workmanship and shall be capable of performing in continuous commercial operation. The Purchaser, who will interpret the meanings of drawings and specification, shall have the power to reject any work or material, which in his judgment is not in accordance therewith. The equipment offered shall be complete with all components necessary for its effective and trouble free operation. Such components shall be deemed to be within the scope of Bidder's supply only, irrespective of whether these are specifically mentioned in his specification and/or the commercial order or not.

1.3 The scope of works envisaged in the specification is detailed below.

(A) <u>Substations:</u>

a) The APTRANSCO shall provide approved Site-cum-Electrical layouts, Control House drawings, Structural & foundation drawings, Bill of materials, other civil & electrical drawings and list of manufacturers/ suppliers for procurement of equipment & materials.

b) The contractor shall execute the works as per the approved drawings mentioned above and shall procure equipment & materials as per specification from approved list of manufacturers.

c) It is the responsibility of the contractor to complete the works in full shape and handover to APTRANSCO in fully operational condition within time specified in the Contract.

(B) <u>Transmission Lines:</u>

a) The APTRANSCO shall provide approved profiles, tower schedules, foundation drawings, Bill of materials, and other technical particulars and list of manufacturers/ suppliers for procurement of materials.

b) The contractor shall procure the materials as per specification from the approved list of manufacturers/ suppliers and execute the works as per the approved profiles. It is the responsibility of the contractor to complete the works in full shape and handover to APTRANSCO in fully operational condition within the time specified in the contract.

c) All the material such as transmission line towers including stubs & cleats, templates, tower accessories, bolts, nuts, spring washers, flat washers, ACSR Moose conductor, disc insulators, HTGS earth wire, hardware & accessories for conductor & earth wire, etc. are to be supplied by the successful bidder.

d) The successful bidder shall execute the line works of conducting check survey and fixing up tower locations as per approved profiles, laying of foundations, erection of towers including accessories, stringing of conductors & earth wire, testing and commissioning of the lines.

2 CLIMATIC CONDITIONS:

2.1. Equipment / material to be supplied against this specification shall be suitable for satisfactory continuous operation under the following tropical conditions:

a)	Maximum ambient air temperature (in shade)	45^0 C
b)	Maximum ambient air temperature (under sun)	50^0 C
c)	Minimum ambient air temperature	7.5^0 C
d)	Maximum daily average ambient air temperature	35^0 C
e)	Maximum yearly average ambient air temperature	30^0 C
f)	Maximum relative humidity	74% to 100 %
g)	Minimum Relative humidity	10%
h)	Altitude above M.S.L.	Up to 1000M
i)	Average No. of thunder storm days per annum	50
j)	Average No. of dust storm days per annum	Occasional
k)	Average No. of rainy days / annum	90
l)	Average Annual Rain fall	925mm
m)	Maximum rainfall per annum	120 Cms
n)	Normal tropical monsoon period	4 months
o)	Maximum wind pressure	150 kg/Sq.M.
p)	Seismic level (horizontal acceleration).	0.1 g

Note: Moderately hot and humid tropical climate conducive to rust and fungus growth. The climatic conditions are also prone to wide variations in the ambient conditions. Smoke is also present in the atmosphere. Heavy lightning also occurs during June to October.

3 QUALITY ASSURANCE PLAN:

The bidder shall invariably furnish the following information along with his offer failing which the offer shall be liable for rejection. Information shall be separately given for individual type of material offered.

i. The structure of organization.

ii. The duties and responsibilities assigned to staff ensuring quality of work.

iii. The system of purchasing, taking delivery and verification of material.

iv. The system for ensuring quality of workmanship.

v. The quality assurance arrangements shall conform to the relevant requirement of ISO-9000, 9002 or 9003 as applicable.

vi. Statement giving list of important raw materials, names of sub-supplies for the raw materials, list of standards according to which the raw material are tested, list of tests normally carried out on raw material in the presence of contractors representative, copies of test certificates.

vii. Information and copies of test certificates in respect of bought out items.

viii. List of manufacturing facilities available.

ix. Level of automation achieved and list of areas where manual processing exists.

x. List of areas in manufacturing process, where stage inspections are normally carried out for quality control and details of such tests and inspections.

xi. List of testing equipment available with the bidder for final testing of material specified and test plant limitation, if any vis-à-vis the type. Special acceptance and routine tests specified in the relevant standards. These limitations shall be very clearly brought out in a Schedule of Deviations from the specified test requirements.

3.1 The Contractor shall within 15 days of placement of order submit the following information to the Purchaser.

i) List of the raw material as well as bought out accessories and the names of sub-suppliers selected from those furnished along with the offer.

ii) Type test certificates of the raw material and bought out accessories if required by the Purchaser / Engineer.

The contractor shall submit the routine test certificates of bought out accessories and central excise passes for raw material at the time of routine testing if required by the Purchaser and ensure that the quality assurance requirements of specification are followed by the sub-contractor.

The Quality Assurance Programme shall give a description of the Quality System and Quality Plans with the following details.

i) Quality System:

- The structure of the organization.
- The duties and responsibilities assigned to staff ensuring quality of work.
- The system of purchasing, taking delivery & verification of materials.
- The system of ensuring of quality workmanship.
- The system of control of documentation.
- The system of retention of records.
- The arrangement of contractor internal auditing.

- A list of administrative and work procedures required to achieve contractors quality requirements. These procedures shall be made readily available to the Purchaser for inspection on request.

ii) Quality Plans:

- An outline of the proposed work and programme sequence.
- The structure of contractors' organizations for the contract.
- The duties and responsibilities ensuring quality of work.
- Hold and notification points.
- Submission of engineering documents required by this specification.
- The inspection of the materials and components on request.
- Reference to contractors' work procedures appropriate to each activity.
- Inspection during fabrication / construction.
- Final inspection and test.

3.2. Training of Personnel:

The Supplier shall provide necessary facilities for training personnel at their works/principals works relating to design, manufacture, assembly and testing and operation and maintenance for six personnel free of cost. However, travel and incidental charges of the personnel will be borne by the purchaser.

4 PACKING AND FORWARDING:

4.1 The equipment shall be packed in crates suitable for vertical/horizontal transport, as the case may be and suitable to withstand handling during transport and outdoor storage during transit. The Bidder shall be responsible for any damage to the equipment during transit, due to improper and inadequate packing and handling. The easily damageable material shall be carefully packed and marked with the appropriate caution symbols. Wherever necessary, proper arrangement for lifting, such as lifting hooks etc., shall be provided. Any material found short inside the packing cases shall be supplied by the Bidder without any extra cost.

4.2 Each consignment shall be accompanied by a detailed packing list containing the following information:

a) Name of the consignee.
b) Details of consignment.
c) Destination.
d) Total weight of consignment.
e) Handling and unpacking instructions.
f) Bill of material indicating contents of each package.

4.3 The Bidder shall ensure that the packing list and bill of material are approved by the Purchaser before dispatch.

5 QUANTITY AND DELIVERY REQUIREMENTS

i) The quantity is mentioned in the price schedules and delivery schedule is set out in Programme of works.
ii) The scope of supply shall include a supply of 2.5% extra quantity of bolts, nuts, washers, split pins, quarter pins and such other loose items free of cost.
iii) The supplier shall arrange for unloading of material at the destination stores/site at their own cost.

6 SUPERVISION SERVICES : NIL

7. MARKING :

All instrument transformers shall have the following nameplate marking on the body for identification.

i) "APTRANSCO "
ii) Purchase Order details
iii) Rating
iv) Tan Delta value at the time of inspection
v) Core details.

8. TYPE TESTS:

All the equipment offered, shall be fully type tested by the bidder as per the relevant standard. The type tests must have been conducted on similar or higher capacity not earlier than **2003**. The bidder shall furnish one set of the following type test reports for each equipment offered along with bid.